

Raw SERS data



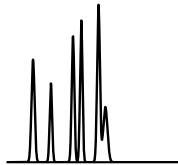
SERS map

one spectrum
per pixel



NNLS gate

background
dropped



pixel spectra x_i

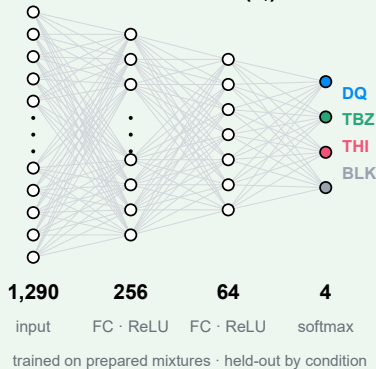
1,290 pts · 500–2500 cm^{-1}



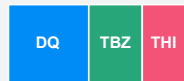
Compositional unmixing

output

shared MLP $f(x_i)$



map composition



per-pixel signature

(bands + composition)

$$\bar{p}_i = \text{pixel mean} \cdot \text{BLK removed}$$

absent component → reported 0

(absence gate, $P < 0.2$)

Concentration

output

reported concentration

nearest measured pixels

weighted avg of MEASURED μM

no match → no answer

declared total (optional)

$$\hat{C}_i = \bar{p}_i \cdot C_{total}$$

overlay when formulation known

library: 92 mixture maps + calibration series